



Walchand Institute of Technology, Solapur
(An Autonomous Institute)

Affiliated to
Punyashlok Ahilyadevi Holkar Solapur University,
Solapur
Choice Based Credit System (CBCS)

Structure and Syllabus
for
Multidisciplinary Minor Program in
Computer Science and Engineering
Offered by
Department of Computer Science and Engineering
W.E.F. 2026-27

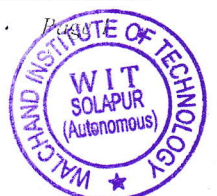
WIT

Dean (Academics)

ARS

HEAD

Computer Sc. and Engg. Deptt
W. I. T., Solapur



Computer Science and Engineering

Vision

To develop professional engineers in Computer Science & Engineering having ethical values, research aptitude and ability to address challenges of modernization in the IT industry aiming at overall sustainable development of the society.

Mission

- M1 - To impart quality education in the field of Computer Science & Engineering in accordance with the needs of the Modernization & Globalization through technology enabled education.
- M2 - To inculcate lifelong learning in students to face challenges posed by ever-changing IT career landscape as a disciplined professional with a sense of professional ethics.
- M3 - To inculcate critical thinking and creativity for identifying various societal issues and to provide solutions.
- M4 - To enhance career opportunities for students through academia-industry interaction and research.

Computer Science and Engineering

Program Educational Objectives (PEOs)

1. Graduate will exhibit strong fundamental knowledge and technical skills in the field of Computer Science & Engineering to pursue successful professional career, higher studies and research.
2. Graduate will exhibit capabilities to understand and resolve various societal issues through their problem solving skills.
3. Graduate will be sensitive to ethical, societal and environmental issues as a software engineering professional and be committed to life-long learning.

Knowledge and Attitude Profile (WK)

WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
WK2	Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
WK5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
WK8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
WK9	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

Program Outcomes (POs)	
PO 1	Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO 2	Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO 3	Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
PO 4	Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
PO 5	Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO 6	The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
PO 7	Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO 8	Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO 9	Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning difference
PO 10	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO 11	Life-long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)
Program Specific Outcomes (PSOs)	
1. Apply the principles of computational mathematics, computer systems and programming paradigms to solve computational problems. 2. Design and develop application software with functionalities applicable for desktop, web and mobile applications with due consideration of system software constraints. 3. Apply software engineering methods, cutting edge technologies and ICT, using appropriate tools and FOSS alternatives for designing, developing & testing application software.	

Computer Science and Engineering Department

Legends Used

L	Lecture Hours / week
T	Tutorial Hours / week
P	Practical Hours / week
FA	Formative Assessment
SA	Summative Assessment
ESE	End Semester Examination
ISE	In Semester Evaluation
ICA	Internal Continuous Assessment
POE	Practical and Oral Exam
OE	Oral Exam
MOOC	Massive Open Online Course
HSS	Humanity and Social Science
NPTEL	National Programme on Technology Enhanced Learning
F.Y.	First Year
S.Y.	Second Year
T.Y.	Third Year
B. Tech.	Bachelor of Technology

Computer Science and Engineering

Course Code Format

2	1	I	T	U/P	2	C	C	1	T/L
Year of Syllabus revision	Program Code	U-Under Graduate P-Post Graduate	Semester No./ Year1/2/3/...8	CourseType e	Course Serial No 1-9	T-Theory, L-Lab session P- Programming			

Program Code

CS	Computer Science and Engineering
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Course Type

BS	Basic Science
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ES	Engineering Science
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HU	Humanities & Social Science
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MC	Mandatory Course
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CC	Core Compulsory Course
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SN*	Self-Learning <i>N* indicates the serial number of electives offered in the respective category</i>
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EN*	Core Elective <i>N* indicates the serial number of electives offered in the respective category</i>
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ON*	Open Elective <i>N* indicates the serial number of electives offered in the respective category</i>
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SK	Skill Based Course
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SM	Seminar
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MP	Mini project
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PR	Project
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IN	Internship
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Sample Course Code

25CSU3CC1T	Discrete Mathematics Structures
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**Multidisciplinary Minor to be offered by
Computer Science and Engineering**

Semester	Name of Course	Engagement Hours			Credits	FA	SA		Total
		L	T	P		Theory	ISE	ICA	
III	Operating System	2	-	-	2	60	40		100
IV	UI Technologies	1	-	2	2		50	25	75
V	Software Engineering	3	-	--	3	60	40		100
VI	Big Data Technologies	2	--	2	3	60	40	25	125
VII	Software Testing and Quality Assurance	3	1	--	4	60	40	25	125
Grand Total		11	1	4	14	240	210	75	525

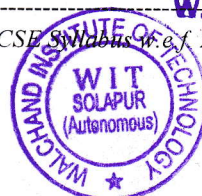
Note:

- N* indicates the serial number of electives offered in the respective category
- ## indicates program code of offering Programme

WIT
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WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
MDM offered by Computer Science and Engineering), Semester-III

25CSU3MD6T - OPERATING SYSTEM

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Credits	2	ISE	40 Marks

Introduction:

This course introduces Fundamentals and basic knowledge of an operating system. It also covers the details Process Management, deadlock, Memory Management and IO subsystems.

Course Prerequisite:

Students should have knowledge of Computer Systems and basics of C programming language.

Course Objectives:

1. To provide basic knowledge about operating systems and processes.
2. To explore process scheduling and deadlock algorithms.
4. To examine memory management techniques for CPU performance.
5. To apply various Unix commands and write shell, awk scripts.

Course Outcomes:

Students will be able to:

1. Comprehend the features of the operating system and process.
2. Analyze process scheduling and deadlock algorithms.
3. Apply memory management techniques for CPU performance.
4. Execute various Unix commands and write shell and awk script.

Unit – I	Introduction	3 Hours
Operating system definition, Simple Batch System, Multi programmed Batch System, Time Sharing System, Parallel System, Real Time System.		
Unit – II	Process	4 Hours
Process Concept, Process Scheduling, Operations on processes, Cooperating Processes, Threads, Inter-Process communication		
Unit – III	Process Scheduling	5 Hours
Basic concept, Scheduling Criteria, Scheduling Algorithms.		
Unit – IV	Deadlocks	6 Hours
System model, Deadlock characterization, Methods for handling deadlocks, Deadlock prevention, Deadlock avoidance, Deadlock detection, Recovery from deadlock.		
Unit – V	Memory Management	5 Hours

Background, Logical Versus Physical Address space, Swapping, Contiguous Allocation, Paging, Segmentation.		
Unit – VI	Virtual Memory	4 Hours
Background, Demand paging, Page replacement, Page replacement algorithms, Allocation of frames, thrashing (Only concept).		
Unit – VII	Introduction to Unix	3 Hours
Unix Commands, Shell Programming, Awk Programming		
Text Books		
1. Operating System concepts, Silberschatz, Galvin, 7th or 8th Edition (John Wiley). 2. Operating Systems: Internals and Design Principles by William Stallings, 5th Edition (PHI). 3. Introduction to Unix Shell Programming, M.G. Venkateshmurthy		
Reference Books		
1. Operating system with case studies in UNIX, Netware and Windows NT by Achyut Godbole (TMGH). 2. Operating Systems, Deitel, Deitel, Choffnes, 3rd Edition, by Pearson Education.		



WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
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MDM offered by Computer Science and Engineering), Semester-IV

25CSU4MD6T – UI Technologies

Teaching Scheme		Examination Scheme	
Lectures	1 Hour/week	ISE	50 Marks
Practical	2 Hours/week	ICA	25 Marks
Credits	2		

Course Prerequisite:

Student should have knowledge of basic programming. They should also have basic knowledge of GUI and its working.

Course Objectives:

1. Inculcate skill necessary to design, develop and style a web-based user Interface.
2. Develop the skill for responsive web design.
3. Develop ability to identify use cases for applying client-side scripting UI.
4. Develop skill required to create light weight browser-based web application using client- side scripting framework

Course Outcomes:

Students will be able to

1. Design, develop and apply styling to a web-based application.
2. To be able to design responsive web design.
3. To develop a web application using different client side/server-side scripting technologies for a given problem.

Unit – I	WEBSITE BASICS, HTML5	4 Hours
Web Essentials: Clients, Servers and Communication – The Internet – World wide web – HTTP Request Message – HTTP Response Message – Web Clients – Web Servers HTML: HTML Review, HTML5 Introduction, Futures of HTML5, New elements in HTML5 (canvas, Media elements, Form elements, Semantic and structural elements)		
Unit – II	CSS	3 Hours
CSS: CSS Introduction, CSS Syntax, CSS Id & Class, CSS Styling, CSS3: Selectors, Box Model Backgrounds and Borders		
Unit – III	CLIENT-SIDE PROGRAMMING- JAVASCRIPT, JSON	4 Hours
Introduction, Understanding of variables, data types, control flow, and basic function usage in JavaScript, Event Handling, JS Built-in Objects JSON: JavaScript Object and Array Creation Using Literals, JavaScript Objects in Arrays & Arrays in Objects, JSON syntax, JSON Parsers, JSON Data Transfer Between Client and Server		
Unit – IV	SERVER-SIDE PROGRAMMING	4 Hours
Servlets: Java Servlet Architecture- Servlet Life Cycle- Form GET and POST, Sessions and cookies		

Internal Continuous Assessment (ICA):

1,2,3 Developing a webpage based on HTML tags
4 CSS
5,6 Applying validation to HTML forms using JavaScript
7,8 JavaScript
9 JSON
10,11 Server-side scripting

Text Books

1. Head First HTML5 Programming by Eric Freeman, Elisabeth Robson O'Reilly Media
2. HTML5 and CSS3, 2nd Edition Level Up with Today's Web Technologies by Brian P. Hogan – Pragmatic Bookshelf, Second Edition.
3. Designing Next Generation Web Projects with CSS3 by Sandro Paganotti- CreateSpace Independent Publishing Platform.
4. JavaScript, A Beginners Guide, Third Edition by John Pollock-McGraw Hill Osborne Media.

Reference Books

1. HTML5 Black Book: Covers CSS3, Javascript, XML, XHTML, Ajax, PHP and JQuery by Kogent Learning Solutions Inc.
2. HTML5 Application, Zachary Kessin, O'Reilly, Shroff Publishing and Distributions Pvt. Ltd.
3. Responsive Design Workflow –Stephen N. Hay.



WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
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MDM offered by Computer Science and Engineering), Semester-V

24CSU5MD6T - SOFTWARE ENGINEERING

Teaching Scheme		Examination Scheme	
Lectures	3 Hours/week	ESE	60 Marks
Credits	3	ISE	40 Marks

Introduction

This course introduces students from diverse engineering disciplines to the principles, practices, and methodologies of Software Engineering. The course emphasizes systematic software development, software project management, software quality assurance, testing, maintenance, and emerging software engineering practices. The knowledge gained will help students understand how software is developed and managed in multidisciplinary engineering projects involving automation, embedded systems, IoT, smart infrastructure, manufacturing, and industrial applications.

Course Prerequisite:

Students should have knowledge of Computer Systems and basics of C programming language.

Course Objectives:

1. Understand the software development process, software engineering principles, and software process models.
2. Apply software engineering techniques for requirements analysis, software design, testing, quality assurance, and project management.
3. Appreciate the importance of software engineering practices in multidisciplinary engineering applications involving automation, electronics, manufacturing, and civil infrastructure.

Course Outcomes:

After successful completion of the course, students will be able to:

1. Explain software engineering principles, software process models, and requirement engineering techniques for systematic software development.
2. Develop software design models, prepare UML diagrams, and apply software testing and quality assurance techniques to engineering applications.
3. Analyze software project management practices and recognize the role of software engineering in multidisciplinary engineering domains such as automation, embedded systems, manufacturing, and civil infrastructure.

Unit – I	Introduction to Software Engineering	7 Hours
Evolution of Software Engineering, Characteristics of Software, Software Crisis, Need for Software Engineering, Software Development Life Cycle (SDLC), Software Process Models : Waterfall Model, Incremental Model, Spiral Model, Prototype Model, Agile Methodology (Introduction) Selection of suitable process model		
Unit – II	Software Requirements Engineering	8 Hours
Requirements Engineering Process, Functional and Non-functional Requirements, Requirement Elicitation Techniques, Software Requirement Specification (SRS), IEEE SRS Standards, Requirement Validation, Introduction to Feasibility Study, Use Case Diagram, Introduction to UML		

Unit – III	Software Design and Modeling	8 Hours
Software Design Principles, Modularity, Coupling and Cohesion, Architectural Design, Structured Design, Object-Oriented Design Concepts, UML Diagrams : Class Diagram, Sequence Diagram, Activity Diagram, State Chart Diagram Design Documentation		
Unit – IV	Software Testing and Quality Assurance	8 Hours
Verification and Validation, Levels of Testing : Unit Testing, Integration Testing, System Testing, Acceptance Testing Black Box Testing, White Box Testing, Test Case Design, Software Quality Concepts, Software Reliability, Software Metrics, Software Reviews and Inspections		
Unit – V	Software Project Management and Emerging Trends	8 Hours
Software Project Planning, Cost Estimation (Basic Concepts), Scheduling using Gantt Chart, Risk Management, Configuration Management, Software Maintenance, Software Documentation, DevOps (Introduction), Version Control Systems : Software Engineering Applications in: Embedded Systems, Smart Manufacturing, Industrial Automation, IoT, Smart Cities, Civil Infrastructure Monitoring		
Unit – VI	Agile Project Management	6 Hours
Introduction to APM, Implementation, Iterative Project Management Life Cycle, Adaptive Project Management Life Cycle, Scrum as a Agile methodology in project management.		
Text Books		
1. Roger S. Pressman and Bruce R. Maxim, Software Engineering: A Practitioner's Approach, 9 th Edition, McGraw-Hill. 2. Ian Sommerville, Software Engineering, 10 th Edition, Pearson. 3. Pankaj Jalote, An Integrated Approach to Software Engineering, 4 th Edition, Springer.		
Reference Books		
1. Rajib Mall, Fundamentals of Software Engineering, 5th Edition, PHI Learning. 2. Richard Fairley, Software Engineering Concepts, McGraw-Hill. 3. Carlo Ghezzi, Mehdi Jazayeri, and Dino Mandrioli, Fundamentals of Software Engineering, Pearson. 4. Watts S. Humphrey, Managing the Software Process, Addison-Wesley.		



WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
MDM offered by Computer Science and Engineering), Semester-VI

24CSU6MD6T -- BIG DATA TECHNOLOGIES

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	4	ICA	25 Marks

Course Objectives:

1. To know the fundamental concepts of big data and analytics
2. To focus on storage, retrieval and processing of big data using Big Data Frameworks like Hadoop, map reduce jobs, Hadoop distributed File system and NOSQL.
3. To study MongoDB CRUD operations and its query language

Course Outcomes:

At the end of this course, students will be able to

1. Describe various formats of digital data, their sources, and the concept of big data.
2. Demonstrate understanding of the concepts of big data analytics and summarize the associated technology landscape.
3. Outline the role of Hadoop in big data processing and illustrate the working of the MapReduce framework.
4. Apply MongoDB concepts to perform CRUD operations on the selected datasets.

Unit – I	Introduction to Types of Digital Data	4 Hours
Classification of Digital Data, Structured Data, Sources of structured data, Ease with Structured data, Semi-Structured data, sources of semi-structured data, Unstructured data, sources of unstructured data, Issues with terminology, Dealing with unstructured data,		
Unit – II	Introduction to Big Data	4 Hours
Big data, What is big data? Why big data? Other characteristics of data which are not definitional traits of big data, Challenges with big data, Big data stack..		
Unit – III	Big Data Analytics	4 Hours
Big Data Analytics, Analytics 1.0, Analytics 2.0, Analytics 3.0, Traditional BI vs. Big Data Environment, Terminologies used in Big Data Environment, Big Data Technology Landscape, NoSQL Databases, NoSQL Vs. RDBMS, NewSQL, Hadoop, Hadoop 1.0 vs. Hadoop 2.0, Data Science is multidisciplinary, Data Scientist		
Unit – IV	Introduction to Hadoop	10 Hours
Introducing Hadoop, Why not RDBMS, Distributed Computing Challenges, A Brief History of Hadoop, Hadoop Overview, Hadoop Components, High Level Architecture of Hadoop, Hadoop Distributed File System, HDFS Architecture, Daemons Related to HDFS, Working with HDFS Command, Special Features of Hadoop.		
Unit – V	Processing Data With Hadoop	4 Hours



Processing Data With Hadoop, Introduction How Map Reduce Works, Map Reduce Example, Word Count Example using Java Managing Resources and Applications with YARN Introduction, Limitation of Hadoop 1.0, Hadoop 2: HDFS, Hadoop 2: YARN, Interacting with Hadoop EcoSystem Hive, Pig, HBase, Sqoop.		
Unit – VI	Introduction to MongoDB	4 Hours
Recap of NoSQL databases, MongoDB – CRUD, MongoDB- Arrays, Java Scripts, Cursors, Map Reduce Programming, Aggregations.		
Internal Continuous Assessment (ICA): - The objective of assignments should be to test students' understanding and assess their ability to put into practice the concepts and terminologies learned. - It should consist of minimum 08 assignments based on following guidelines - Basic operations of RDBMS (DDL, and DML) - Basic big data operations using NumPy, SciPy & Pandas. - Implementation of Plotting a CSV File Data using Pandas. - Implementation of Filtering and Cleaning a CSV File Data Using NumPy & Pandas. - Hadoop basics and programming - Map Reduce Programming - Data manipulation operations on SQL, NoSQL and NewSQL. - Simple MongoDB and its CRUD Operations - MongoDB Arrays, cursors - Performing import, export and aggregation in MongoDB.		
Text Books		
- Big Data and Analytics, Seema Acharya, Subhashini Chellappan, - Wiley India Pvt. Ltd. - Hadoop: The Definitive Guide, 3rd Edition, Tom White, - O'reilly Media. - The Definitive Guide to MongoDB: A Complete Guide to Dealing with Big Data Using MongoDB (Definitive Guide Apress) 2e by David Hows, Eelco Plugge, Peter Membrey, Tim Hawkins.		
Reference Books		
- Big Data For Dummies, Judith Hurwitz, Alan Nugent, Dr. Fern Halper, Marcia Kaufman, Wiley Brand. - Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends for Today's Businesses (Wiley CIO), Michael Minelli, Michele Chambers, Ambiga Dhiraj: John Wiley & Sons. - Mining of Massive Datasets, Anand Rajaraman, Jure Leskovec, Jeff rey D. Ullman, Cambridge University Press. - Hadoop in Action, Chuck Lam, Dreamtech Press, ISBN : 978-81-7722-813-7.		

WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
B.Tech. (Computer Science and Engineering), Semester-VII

24CSU7MD5T- Software Testing and Quality Assurance

Teaching Scheme		Examination Scheme	
Lectures	3 Hours/week	ESE	60 Marks
Tutorials	1 Hour/week	ISE	40 Marks
Credits	4	ICA	25 Marks

Course Objectives:

1. To describe knowledge of the software testing process and various methods of testing.
2. To describe generation and execution of test strategy & test plan.
3. To discover correctness, completeness and quality of software.
4. To implement automated testing tools to test software.

Course Outcomes:

At the end of this course, students will be able to

1. Compare the different software testing methods and select the suitable one for a given scenario.
2. Design test strategy & test plan for software testing.
3. Apply different approaches of quality assurance and standards for increasing quality of software.
4. Demonstrate automated testing tools to test software.

Unit – I	Fundamentals of Software Testing	6 Hours
Introduction, Basics of Software Testing, Approaches to Testing, Testing During Development Life Cycle, Essential of Software Testing, Features of Testing, Misconceptions About Testing, Principles of Software Testing, Waterfall Model and Agile Testing Fundamentals, Test Policy, Strategy, Planning, Process, Challenges in Testing, Test Team Approach, Methods, Defect Classification, Defect, Error, Mistake in Software, Defect Life Cycle, Defect Management Process, Developing Test Strategy, Test Methodologies, Skills Required by Tester.		
Unit – II	Methods of Testing	6 Hours
Software Verification and Validation, Black-Box and White-Box Testing, Static and Dynamic Testing, Black-Box Testing Techniques-Equivalence Partitioning, Data Testing, State Testing, Other Black Box Test Techniques. White-Box Testing Techniques-Data Coverage, Code Coverage, Other White Box Test Techniques.		
Unit – III	Levels of Testing	6 Hours
Verification and Validation Model, Levels of Testing, Requirement Testing, Design Testing, Code Review, Unit Testing, Module Testing, Integration Testing, Big-Bang Testing, Sandwich Testing, System Testing- GUI Testing, Compatibility Testing, Security Testing, Performance Testing, Volume Testing, Stress Testing, Load Testing, Installation Testing, Regression Testing, Smoke Testing, Sanity Testing, Ad hoc Testing, Usability Testing, Acceptance Testing-Alpha Testing, Beta Testing, Gamma Testing.		

Unit – IV	Test Planning & Documentation	8 Hours
Test Planning-The goal of Test Planning, Test Planning Topics, Writing and Tracking Test Cases-The Goal of Test Case Planning, Overview, Test Case Organization and Tracking, Reporting Bugs- Getting Your Bugs Fixed, Isolating and Reproducing Bugs, Bug-Tracking Systems.		
Unit – V	Test Metrics	4 Hours
Test Measurements, Test Metrics, Metric Life Cycle, Types of Manual Test Metrics.		
Unit – VI	Quality Concepts & Software Quality Assurance	8 Hours
Quality Concepts-What is Quality?, Software Quality, The Software Quality Dilemma, Achieving Software Quality, Elements of Software Quality Assurance, SQA Processes and Product Characteristics, SQA Tasks, Statistical SQA, Software Reliability, CMM, The SQA Plan.		
Unit – VII	Automated Testing and Testing Tools	6 Hours
Introduction, The Benefits of Automation and Tools, Software Test Automation, Random Testing, Realities of Using Test Tools and Automation.		
Internal Continuous Assessment (ICA): 1. Understand the requirement, generate the test case, analyse the bug using ChatGPT AI tool. 2. Develop a comprehensive test plan for a simple application (e.g., login system or calculator etc). 3. Use Claude AI for designing test strategy and complex test scenarios. 4. Understand and Apply black-box techniques to a sample problem. 5. Generate a basic automated test using Selenium Web Driver or Cypress. 6. Convert Manual test cases into automation script (Python / Java) using Copilot AI tool. 7. Use cursor AI tool and generate selenium for different scenarios. 8. Identify regression test cases for a feature update in an existing application using ChatGPT AI tool. 9. Perform API testing using POSTMAN tool. 10. Perform performance testing for a web application Using JMeter or LoadRunner.		
Text Books		
1. Software Testing Principles, Techniques and Tools By M G Limaye, Published by Tata McGraw-Hill Education Private Limited, Published 2009, ISBN (13): 978-0-07-013990-9, ISBN (10): 0-07-013990-3 (Chapter 1 & 3) 2. Software Testing, Second Edition By: Ron Patton, Published by SAMS, ISBN-13: 978-0672327988 ISBN-10: 0672327988 (Chapter 2, 4 & 6) 3. Software Engineering: A Practitioner's Approach by Roger S Pressman, 8th Edition, Publisher McGraw Hill (Chapter 5)		

Reference Books

1. Software Testing Principle and Practices By Ramesh Desikan, Gopalaswamy Ramesh, Pearson Education, ISBN 978-81-7758-121-8
2. Software Testing Principles and Practices By Naresh Chauhan, Publisher OXFORD UNIVERSITY PRESS-NEW DELHI, ISBN 0-19-806184-6
3. Beautiful Testing: Leading Professionals Reveal How They Improve Software By Adam Goucher, Tim Riley, Publisher O'reilly
4. Foundations of Software Testing By Rex Black, Dorothy Graham, Erik Van Veenendaal, Isabel Evans, Published by Cengage Learning India Pvt Ltd.
5. Lessons Learned in Software Testing by CemKaner, James Bach, Bret Pettichord, Publisher Wiley
6. Testing Computer Software CemKaner, Jack Falk, Hung Q. Nguyen, Publisher Wiley
Selenium Testing Tools Cookbook By Unmesh Gundecha Published by Packt, ISBN: 978-1-84951-574-0
7. Dr. K.V.K.K. Prasad, "Software Testing Tools: Covering WinRunner, Silk Test, LoadRunner, JMeter and TestDirector With Case Studies", Dreamtech Publications, ISBN: 10:81-7722-532-4